Effective: 3 October 2024
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PROCEDURES ON RISK BASED PRICING MODEL (RBPM) FOR BANKING FACILITIES

Version 5 (2024)

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APENDICES

Appendix A - List of Internal Policies/Procedures and Documents issued by relevant authorities
Appendix B - Calculation of spread

Any updates to this Procedures shall be recorded in the following table:

Circular No.	Date	Page No	Description	Approved by
CG/2008/18	30/07/2008	All	Version 1.0 (2008)	8th Management Risk Committee (MRC) on 30th July 2008
CG/2015/48	08/09/2015	All	Version 2.0 (2015)	104th Management Risk Committee (MRC) on 17th August 2015
CG/2019/89	1/10/2019	All	Review Exercise Version 3.0 (2019) - Update on operational matters subsequent to the latest review on the pricing models. - Chapter 2 (Loan and Financing Pricing) - Chapter 3 (Roles and Responsibilities) Chapter 4 (Process)	151st Management Risk Committee (MRC) on 22nd August 2019.
CG/2023/06	9/2/2023		Revised Version 4 (2023) -updated to be in line with the revised RBPM methodology approved in 315th BOD held on 30th May 2022. -updated the new department/ division/ Committees' name.	35th Management, Risk and Compliance Committee meeting 11 January 2023.
CG/2024/85	3/10/2024		Revised Version 5 (2024) -updated to be in line with the revised RBPM	55th Management, Risk and Compliance

			methodology approved in 334th BOD held on 08th February 2024. -updated the new department/ division/ Committees' name.	Committee meeting on 19 September 2024

Note: *For more information about this document updates as stated above, please refer to the Market Risk Department.*

Name	Description
ALCO	Asset Liabilities Committee
AUD	Australian Dollar
ARRD	Asset Rehabilitation & Recovery Department
BNM	Bank Negara Malaysia
BOD	Board of Directors
BRC	Board Risk Committee
COF	Cost of Fund
CRD	Credit Risk Department
DFI	Development Financial Institution
ECL	Expected Credit Loss
EURIBOR	Euro Interbank Offered Rate
EUR	EUR Currency
EXIM Bank	Export-Import Bank of Malaysia
GBP	British Pound Currency
KLIBOR	Kuala Lumpur Interbank Offered Rate
KLIRR	Kuala Lumpur Islamic Reference Rate
LAF	Loans, Advances and Financing
LGD	Loss Given Default
MRCC	Management Risk and Compliance Committee
MYR	Malaysian Ringgit Currency
PD	Probability of Default
RMD	Risk Management Division
RBPM	Risk Based Pricing Model
SGD	Singapore Dollar Currency
SIBOR	Singapore Interbank Offered Rate
SOFR	Secured Overnight Financing Rate
MRD	Market Risk Department
USD	US Dollar Currency

Terms	Definition
Bank/EXIM Bank	Export-Import Bank of Malaysia Berhad
This Document	Refers to Procedures on RBPM for Banking Facilities

(A) **Background**	The Document has been approved for compliance by all staff of Export-Import Bank of Malaysia Berhad ("EXIM Bank") involved either directly or indirectly in the Procedures on Risk Based Pricing Model for Banking Facilities. This Document prescribed the concept and methodology related to the Risk-Based Pricing Model (RBPM) comprises of Cost of Fund (COF) and Spread. It is intended to ensure the appropriateness and standardize approaches in evaluating and determining the loan and financing pricing for banking facilities based on the concept of risk-return relationship. This Document will be circulated to all staff of EXIM Bank.
(B) **Document Owner**	The Market Risk Department ("MRD"), Risk Management Division ("RMD") is responsible for the development, updates and review of the Document.
(C) **Interpretation of the Document**	In the event of any ambiguity on the content or where the interpretation of this Document is required, the Head of MRD must be consulted.
(D) **Other Policies & Procedures**	This Document must be read together with other related internal policies and/or Procedures issued by EXIM Bank. Please refer to Appendix A for further details.
(E) **Compliance with Laws and Regulations**	This Document is to be read together with any written laws as at the material time enforced in Malaysia and/or guidelines, policies and circulars issued by any relevant authorities such as BNM. In the event of any contradiction between this document with the laws and/or guidelines, policies and circulars issued by any relevant authorities, the latter must prevail. Please refer to Appendix A for further details.
(F) **Anti-Bribery and Corruption Commitments**	EXIM Bank has developed and implemented controls to ensure commitment from business associates to prevent bribery and corruption. EXIM Bank will terminate the relationship with business associate in the event of bribery or corruption by, on behalf of, or for the benefit of the business associate in connection with the relevant transaction, project, activity, rendered services or relationship.
(G) **Exception to this Document**	Any deviation from this Documents must be approved by the MRCC.

(H) Infringement of the Document	Any infringement of this Procedures shall constitute as non-compliance and warrants a disciplinary action against the offender.
(I) Review of the Document	This Document shall be reviewed once in 2 years or as and when required.
(J) Getting Help	Any query pertaining to the content of this document or request for further clarification can be forwarded to the Head, MRD.
(K) Change Request	Any recommendation for amendments or changes to this document must be forwarded to the Market Risk Department using the Document Change Request Form available from GPMD.

(A)

General

1) The pricing of loans and financing is a key element in risk management to ensure the Bank's exposures and its activities are adequately compensated, as well as mitigating a commercially driven pricing to ensure the Bank's soundness and long-term business sustainability.

2) All relevant departments that involve directly or indirectly in granting loans and financing facilities to the borrower/customer are required to understand thoroughly the Procedures and to ensure its compliance at all times. Any violation must be treated as a breach of discipline and therefore subject to the disciplinary action.

3) In the absence of this written Procedures or when it doubts, the appropriate authority shall be consulted accordingly.

(B)

Objective

The objective of the Procedures is to facilitate the following:

1) To have a standard method and approach in the evaluation and determination of the interest/profit rates of loans/financing of the Bank which reflects a risk return relationship.

2) To provide a clear direction and guidance to all staff of EXIM Bank involved either directly or indirectly in assessing the `loan and financing pricing for banking facilities.

3) To have an effective monitoring on the interest/profit rates assessment based on the Loan/Financing Pricing; and

4) To ensure it complies with the prescribed Loan/Financing Pricing for the interest/profit rates to be charged to the borrower/customer.

(C)

Effective Date

All staff are required to comply with the requirements of this Procedures with immediate effect.

(D)

Coverage

The Procedures is applicable for Loans, Advances and Financing (LAF) activities which are extended to both Conventional and Islamic facilities.

(A) Introduction

1) This RBPM set the profit/interest rates for the Bank's loans and financing facilities extended to both conventional and Islamic banking.
2) Its implementation shall reflect risk-return relationship in the interest/profit rates charged to the borrower/customer based on the type of facilities (trade, non-trade or syndication), Probability of Default (PD Rating), Loss Given Default (LGD Score) and tenure.

(B) Methodology for RBPM Components

1) RBPM consist of two major components, the cost of fund (COF) and spread.

Under each major component, it is further breakdown into separate pricing components that represent different factors in pricing. The components were last reviewed in April 2024 and the details of the methodology are as follows:

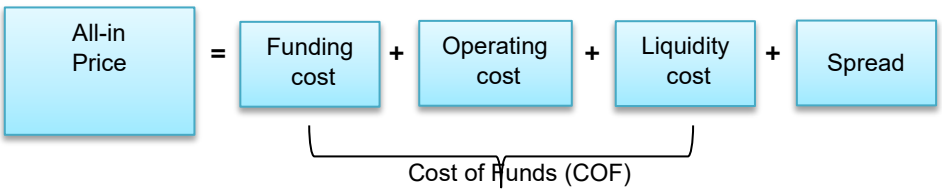


Diagram 1: Component of Risk Based pricing Model (RBPM)

Funding cost	a. Reference rate	An interest rate benchmark depending on payment frequency.
	b. Funding spread	Weighted average of the actual borrowing/funding cost over the reference rate.
Operating cost		Operating expenses as a percentage of total exposure cost
Liquidity cost		The cost associated with the holdings of high-quality liquid assets (HQLA)
Expected credit lost (ECL)		The cost that the Bank is expected to incur in the event of default.

Spread	1) A minimum spread has been implemented to establish a chargeable margin for a customer. 2) Tenor spread has been introduced, from less than 1-year (0%) to more than 10-year (1.15%) respectively. Applicable for fixed rates only.

*Detailed of the calculation is as per Appendix B.

(C)

Cost of Fund (COF) for floating based facilities

1) COF for the floating based facilities comprises of funding cost, operating cost and liquidity cost.

2) Funding cost consists of (i) reference rate plus (ii) funding spread.

3) Reference rate is an interest rates benchmark for the key currencies i.e., MYR, USD, GBP, EUR, SGD and AUD.

4) The reference rate covers significant tenure i.e., 1 month,3 month, 6 month, 9 month, 12 month

5) The reference rate shall follow the Alternative Reference Rate (ARR) once the ARR comes into effect.

6) Funding spread represents the Bank's actual borrowing/funding cost that is charged by financier. Computation of the funding spread is determined as follows:

   a) Trade facilities

      i. MYR

      Based on weighted average of the actual short-term borrowing and financing cost. Government initiatives e.g., Teraju, BNM SME Program are eligible for subsidy rate

      ii. USD

      Based on weighted average actual short-term borrowing and financing spread charged by lender/financier for both conventional and Islamic funding.

iii. Other than USD i.e. EUR, GBP, AUD and SGD

To be quoted by Treasury Department. The actual COF charged by the funder will be transferred (back-to-back) to the specific customer funded by the respective source.

b) Non-Trade Facilities

i. MYR

Based on weighted average of the actual long-term borrowing and financing cost. Government initiatives e.g., Teraju, BNM SME Program are eligible for subsidy rate

ii. USD

Based on weighted average actual long-term borrowing and financing spread charged by lender/financier for both conventional and Islamic funding.

iii. Other than USD i.e., EUR, GBP, AUD and SGD

To be quoted by Treasury Department. The actual COF charged by funder will be transferred (back-to-back) to the specific customer funded by the respective fund.

c) Syndication

i. USD

Based on weighted average actual long term and short-term borrowing and financing spread charged by lender/financier for both conventional and Islamic funding.

ii. Other than USD i.e., EUR, GBP, AUD and SGD

To be quoted by Treasury Department. The actual COF charged by funder will be transferred (back-to-back) to the specific customer funded by the respective fund.

7) Operating cost

Methodology	Computation
Operating expenses as a percentage of total exposure	$\frac{\text{Operating expenses}}{\text{Total assets (Exclude LAF) +Gross Loans and Financing + Commitments}}$

a) To align both the operating expenses and total exposure on a complete basis.

- Operating cost includes personnel cost, establishment and marketing expenses, and general administrative expenses.
- The insurance cost is excluded from the computation as insurance will have its own RBPM calculation.

b) The Operating Cost computation will be based on the projection calculation for the year 2024.

8) Liquidity cost

a) The costs associated with the holding of High-Quality Liquid Assets (HQLA) to ensure the Bank is able to meet its liquidity requirement.

b) Calculated based on Weighted Average Coupon for Non-HQLA minus Weighted Average Coupon for HQLA over Earning Assets.

c) This will be subjected to regular review and updated accordingly.

(D) Expected Credit loss (ECL) Cost

1) ECL Cost is the cost that the Bank is expected to incur in the event of default

2) ECL Cost is calculated as the Probability of Default (PD) multiplied Loss Given Default (LGD) for each Exposure at Default (EAD).

3) The ECL cost is subjected to PD rating and collateral which is industry cost competitive.

4) The PD will be based on the risk rating generated by Credit Risk Management System (CRMS).

5) LGD will be based on the Bank's historical data that is further segmented to Fully Secured, Partial Secured and Unsecured.

6) The LGD score is based on the historical recovery rate of the Bank.

7) The ECL can be reconsidered at where in terms of pricing, only 50% of the ECL cost will be reflected if Banking Division demonstrate details and specific justification in their credit paper with one of the following:

   a. The new credits onboarded will have to be acceptable credit quality in line with the Bank's target market risk acceptance credit risk and subject to robust credit and legal assessment.
   b. The new credits onboarded are aligned to the Bank's mandate as an Exim Bank to support the country strategic export import sector as well as to provide financing to the underserved segment.
   c. Justification based on the BNM's Performance Measurement Framework where "additionality" must demonstrate e.g. new job creation, industry backward linkages, social economic impact, development of strategic sectors (agriculture, infrastructure, export, capital intensive and high technology), export of Malaysian expertise/products to overseas; and promotion of Malaysian companies' skillsets/expertise/brand in the international arena.

(E) Spread

1) For floating rate, a minimum spread has been implemented to establish a chargeable margin for a customer. The calculation is as follows:

   ECL Cost + Profitability Margin

2) For fixed rate, tenor spread has been introduced, from less than 1-year (0%) to more than 10-year (1.15%) respectively. Applicable for fixed rates only. The calculation is as follows:

   ECL Cost + Profitability Margin + Fixed rate spread

4) The Business Division must adhere to the minimum spread outlined in the tables provided in Appendix C, specifically Appendices 3a and 3b.

(F)

Applicability

1) Cost of Fund (COF)

a) Cost of Fund (COF) is compulsory above the reference rate and shall be applied based on currency and facilities type of the loans, advances and financing which is extended to both conventional and Islamic banking.

b) COF shall be charged to borrower/customers based on the following:
   i. Type of facilities – trade, non-trade, syndication, floating-based or fixed-based facilities.
   ii. Currency – MYR, USD, EUR, GBP, AUD and SGD.

c) COF for floating-based facilities to be charged to borrower/customers shall subject to changes based on the periodical review by the Bank. However, its position for fixed-based facilities shall remain until the maturity.

d) Reference rate shall be determined as follows:
   i. Type of currency – KLIBOR, KLIRR, SOFR, EURIBOR, SOR/SIBOR and AUD-BBSW.
   ii. Payment frequency – 1 month, 3 months, 6 months, 9 months and 12 months.

e) Reference rate shall change based on its prevailing rates following the payment date of loans, advances and financing.

2) Spread

a) Actual spread (as proposed in the loans/financing proposal) shall be charged to borrowers/customers subject to approval by the relevant approving authority.

b) Any override/change of RBPM's Spread/Margin, it must be supported with appropriate justification in the relevant management committee.

c) Spread that is charged to the borrowers/customers may be changed subject to annual review of the account or significant changes to the PD rating or LGD scored of the borrowers/customers.

(A) **Board of Directors (BOD)**	The BOD is responsible for approving any changes to the RBPM i.e., Cost of Fund (COF) and Spread, and its implementation.
(B) **Board Risk Committees (BRC)**	The BRC is responsible for: a) recommending approval of any changes to the RBPM tabled by the Market Risk Department. b) overseeing the implementation of the RBPM within the Bank.
(C) **Shariah Committee (SC)**	The SC is responsible to endorse any changes in the Procedures on Risk Based Pricing Model for Banking Facilities particularly on matters related to Islamic Financing to mitigate Shariah Non-Compliance Risk.
(D) **Management Risk and Compliance Committee (MRCC)**	MRCC is the approving authority for any changes to the Procedures on Risk Based Pricing Model for Banking Facilities.
(E) **Asset Liabilities Committees (ALCO)**	ALCO is responsible for: a) reviewing any changes to the RBPM prior tabling to the BRC and BOD for approval. b) monitoring the implementation of the Procedures.
(F) **Management Credit Committee (MCC)**	MCC responsibilities include: a) the final approving authority for the proposed interest/profit rates to be charged to the borrowers/customers. b) granting approval on a case-to-case-basis should lower all in cost rate to be charged to the borrowers/customers as prescribed in the RBPM.
(G) **Market Risk Department (MRD)**	MRD responsibilities include: a) the owner of the RBPM and its Procedures. b) any clarification or updates to the RBPM and its Procedures. c) to report on yield's performance on loans, advances and financing upon the implementation of the Procedures. d) to ensure proper communication and dissemination of the Procedures to various stakeholders.

	e) to highlight issues and recommendations related to the RBPM and the Procedures.
(H) **Treasury Department (TD)**	TD is responsible for: a) the ownership and overall review of the Cost of Fund (COF). b) communicate with the Banking Division in relation to COF and the reference rates. c) notify the Asset Liabilities Committees (ALCO) if there is a significant change in the Bank's actual borrowing and financing costs prior to the periodical review periods.
(I) **Banking Division (BD)**	BD is responsible to: a) ensure the interest/profit rates to be charged to borrowers/customers as per Procedures on Risk Based Pricing Model for Banking Facilities. b) ensure that the interest/profit rates is charged to borrowers and to obtain approval from the MCC (on a case-to-case basis) should a lower rate be charged. c) to inform Risk Management Division and other relevant departments on the final profit rate charged to the borrowers/customer that has been approved. d) to provide information and recommendations for the review of RBPM and the Procedures.
(J) **Credit Risk Department (CRD)**	CRD is responsible to: a) advise on the proposed spread based on the final PD rating and LGD score of the customers. b) highlight if there is pricing deviation in their assessment for MCC's approval.

(A)

Loan Pricing Determination

1) Banking Division (BD)

a. BD must propose the interest/profit rate to be charged to the borrower/customer based on the initial PD rating. However, the final interest/profit rate is subjected to the final PD rating and LGD score by the Credit Risk Department.

b. the final interest/profit rate to be charged to the borrower/customer shall be subject to approval by the MCC.

c. BD must seek approval from the MCC should the proposed interest/profit rate to be charged is lower than the prescribed interest rate.

2) Treasury Department (TD)

a. TD must distribute the prevailing reference rates i.e., KLIBOR, KLIRR*, SOFR, EURIBOR, SIBOR/SOR and AUD-BBSW to the BD, Credit Operation Department and Market Risk Department on daily basis, as well as the COF for all currencies.

***** With the discontinuation of KLIRR, BNM SAC ruling (218 th SAC Meeting dated 28 October 2021) allows temporary referencing to conventional term reference rate such as KLIBOR or other conventional term reference rates on Hajah basis, until a reliable Islamic term reference rate is available. This is to prevent any disruption to the market given the relatively lower liquidity in the Islamic market.

(B)

Review Cost of Fund (COF)

1) Market Risk Department (MRD)

a. MRD must conduct independent verification on review of the COF and recommends for approval by ALCO**.**

2) Treasury Department

a. TD is responsible for overall review of the Cost of Fund (COF) on **a semi-annual basis**.

b. TD must distribute the final review of COF to the Banking Division and Credit Operation Department together with the latest reference rate based on the respective currencies.

(C)
Monitoring

1) Market Risk Department (MRD)

a. MRD shall monitor the final interest/ profit rates to be charged to borrowers/customers as approved by the respective approving authority on **monthly basis** and shall be reported to Chief Risk Officer, Risk Management Division.

b. MRD will notify ALCO and Board Risk Committee (BRC) on price/yield movement on monthly basis.

List of Internal Policies/Procedures and Documents issued by relevant authorities

Internal Policies/Procedures issued by EXIM Bank

1. Risk Appetite Policy version 1 (2022)
2. Procedures on Policy, Procedures and Standard Operating Procedures (PPS) Formulation and Review Version 5 (2023)
3. Risk Management Framework, Version 3 (2023)
4. Credit Risk Policy
5. Approving Authorities and Authority Limits Version 10(2024)

Documents issued by relevant authorities

1. Guidelines on Best Practices for the Management of Credit Risk for Development Financial Institutions (27 July 2009)
2. Credit Risk (31 July 2023)
3. Credit Risk – Best Practices for Development Financial Institutions (DFIs) (7 June 2018)

Appendix 3a: Minimum Spread

Minimum Spread = ECL Cost + Profitability Margin.

Profitability margin defined as a Projected Profit Before Tax for Year 2024 (PBT) as a percentage of Projected Total Assets for Year 2024.

$$\text{Profitability Margin} = \frac{\text{Projected Profit Before Tax Year 2024 (PBT)}}{\text{Projected Total Assets Year 2024}}$$

$$\text{Profitability Margin} = \frac{38{,}557}{8{,}155{,}832} \times 100 = \mathbf{0.4728\%}$$

ECL Cost to be charged at 100%. However, 50% charge is allowed provided ***one*** of the ***3 conditions*** is fulfilled (refer to Appendix 1(4)- ECL Cost).

Table 1: 100% ECL

PD	Fully Secured	Partially Secured	Unsecured
1	0.0001%	0.0059%	0.0121%
2	0.0004%	0.0178%	0.0364%
3	0.0016%	0.0705%	0.1442%
4	0.0060%	0.2656%	0.5428%
5	0.0211%	0.9369%	1.9150%
6	0.0588%	2.6111%	5.3372%
7	0.3164%	14.0573%	28.7331%
8	0.8900%	39.5400%	80.8200%
9	0.8900%	39.5400%	80.8200%

Table 2: 50% ECL

PD	Fully Secured	Partially Secured	Unsecured
1	0.0001%	0.0030%	0.0061%
2	0.0002%	0.0089%	0.0182%
3	0.0008%	0.0353%	0.0721%
4	0.0030%	0.1328%	0.2714%
5	0.0106%	0.4685%	0.9575%
6	0.0294%	1.3056%	2.6686%
7	0.1582%	7.0287%	14.3666%
8	0.4450%	19.7700%	40.4100%
9	0.4450%	19.7700%	40.4100%

Table 3: Minimum Spread based on 100% ECL

PD	Fully Secured	Partially Secured	Unsecured
1	0.473%	0.479%	0.485%
2	0.473%	0.491%	0.509%
3	0.474%	0.543%	0.617%
4	0.479%	0.738%	1.016%
5	0.494%	1.410%	2.388%
6	0.532%	3.084%	5.810%
7	0.789%	14.530%	29.206%
8	1.363%	40.013%	81.293%
9	1.363%	40.013%	81.293%

Table 4: Minimum Spread based on 50% ECL

PD	Fully Secured	Partially Secured	Unsecured
1	0.473%	0.476%	0.479%
2	0.473%	0.482%	0.491%
3	0.474%	0.508%	0.545%
4	0.476%	0.606%	0.744%
5	0.483%	0.941%	1.430%
6	0.502%	1.778%	3.141%
7	0.631%	7.501%	14.839%
8	0.918%	20.243%	40.883%
9	0.918%	20.243%	40.883%

Appendix 3b: Tenor spread for Fixed Rate Loan only.

The spread is an average of multiple yield of MGS bonds* minus Overnight Policy Rate (OPR**) that taken from Bloomberg, which the breakdown of each tenor's spread is shown below:

Tenure	Proposed additional tenor spread to the minimum spread (refer Appendix 3a)
< 1 year	0.0000%
1 year - 3 years	0.4510%
3 years – 5 years	0.6037%
5 years – 10 years	0.8311%
> 10 years	1.1488%

** Source: Average multiple yields of MGS bonds are taken from Bloomberg, as of 29 September 2023.*

***OPR rate 3.00% as of 29 September 2023.*

The above tenure risk should only be priced on fixed rate loans given floating facilities are subject to 3 to 6 months reset/reprice of the floating rate.


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